

ToyJoy Inventory Optimization Analysis

01. Executive Summary

ToyJoy sells toys through retail and online channels and faced major inventory planning issues after recent holiday seasons. Some products remained heavily overstocked, tying up warehouse capacity and cash, while high-demand products frequently ran out of stock, causing missed sales opportunities. This analysis evaluates sales, inventory levels, and seasonal demand patterns over the last three years to identify inventory inefficiencies and improve stock planning.

02. Business Problem

ToyJoy currently lacks clear visibility into:

- Which products are overstocked and tying up capital
 - Which products frequently go out of stock
 - How demand changes during holidays and school seasons
 - How inventory should be adjusted for high-demand and underperforming products
 - How reorder planning can reduce both overstock and stockouts
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03. Business Objectives

- Identify products and categories with excess inventory
 - Detect products with frequent stockout risk
 - Analyze seasonal demand patterns over the last 3 years
 - Recommend inventory adjustments for slow-moving and high-demand items
 - Improve reorder planning and stock efficiency
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04. Dataset Overview

The dataset contains inventory and sales records for ToyJoy across multiple stores over a 3-year period from January 2023 to December 2025. Each row captures product-level sales, inventory position, unit pricing, unit cost, and seasonality information. The dataset is designed to evaluate inventory performance, identify stock risks, and understand how seasonal demand affects stocking decisions.

05. Overall Business Performance

KPI SUMMARY

87M Total Revenue	176K Total Units Sold	85M Total Inventory Value
2.28K Inventory Turnover Rate	3K Overstock Products Count	118 Stockout Risk Count

Revenue Performance

ToyJoy generated approximately **₹87M in total revenue over the 3-year analysis** period, indicating strong overall sales performance across its toy portfolio and retail network.

Sales Performance

The company sold **approximately 176K units**, showing healthy customer demand across multiple toy categories, especially during peak seasonal periods such as holidays and school-related demand cycles.

Inventory Value

ToyJoy is currently holding approximately **₹85M worth of inventory**, which is very close to its total revenue. **This indicates that a significant amount of working capital is tied up in stock and requires better inventory balancing.**

Inventory Turnover Performance

The inventory turnover rate of 2.28K suggests that products are moving through inventory at varying speeds, with some products selling efficiently while others remain in stock for longer periods.

- 2 to 4 → slow/moderate moving inventory
- 4 to 8 → healthy / good for many retail businesses
- 8+ → very fast moving inventory
- below 2 → possible overstock / slow-moving stock

Overstock Performance

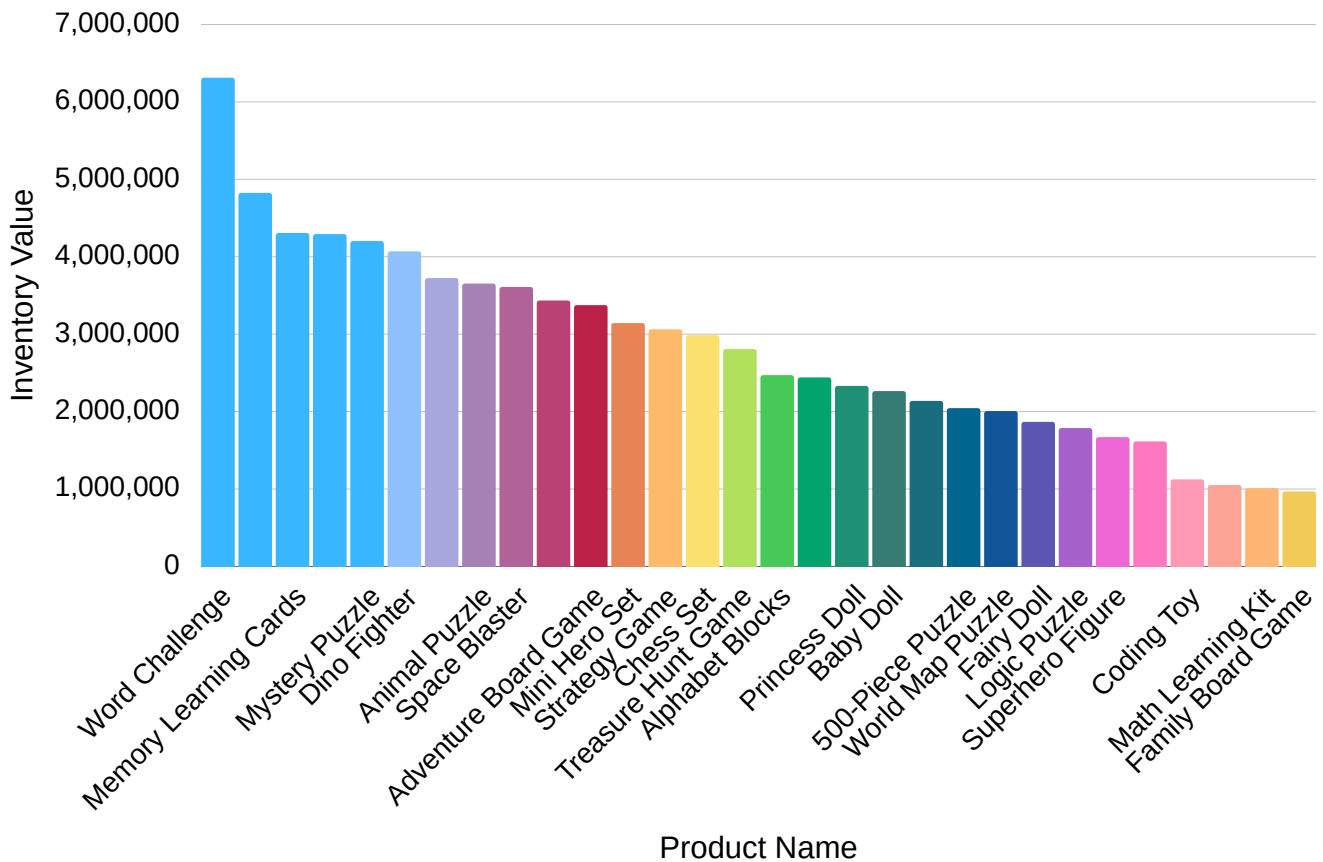
The analysis identified around **3K overstock product records**, highlighting that several products are carrying excess inventory beyond actual demand. This overstock ties up warehouse space and cash flow.

Stockout Risk Performance

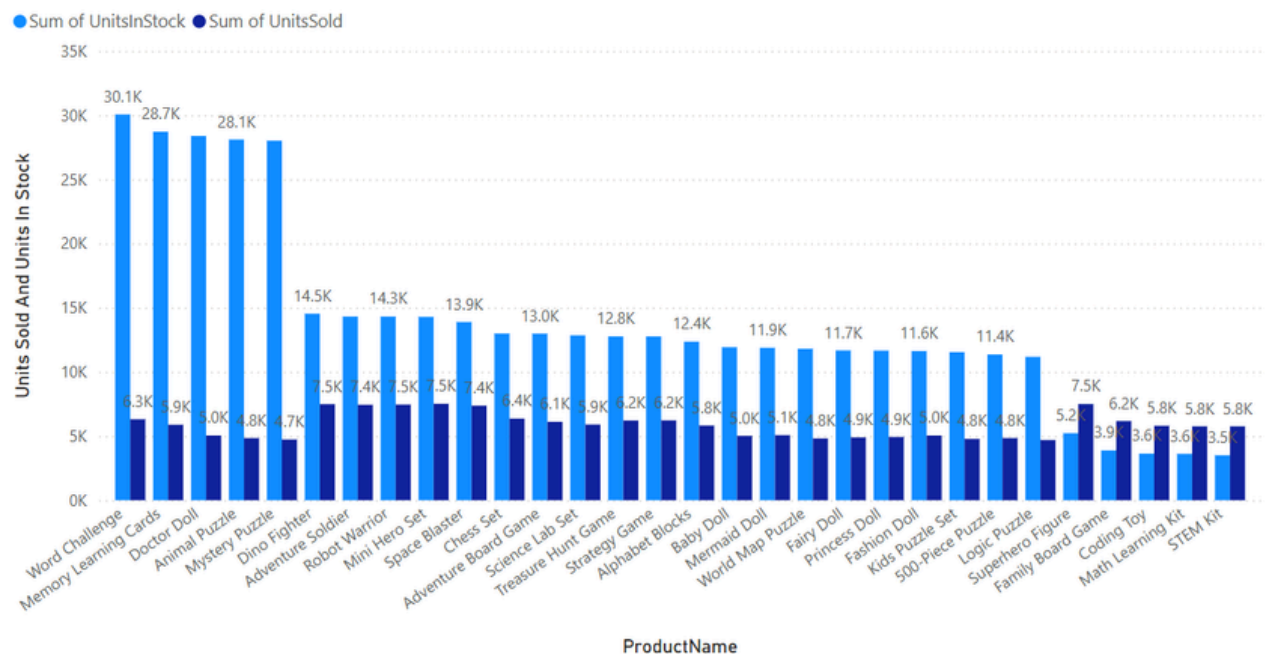
The dataset also shows **118 stockout-risk cases**, indicating that some high-demand products frequently approach zero inventory levels, which may lead to lost sales and reduced customer satisfaction.

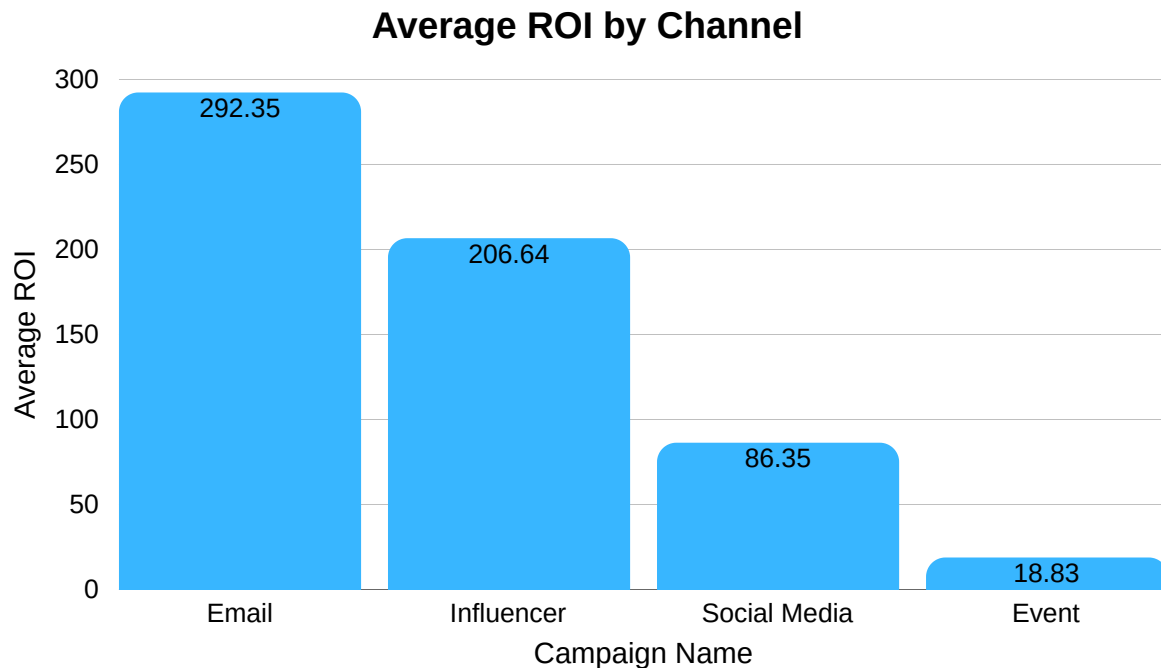
06. Excess Inventory Analysis

Inventory Value by Product



Units Sold vs Units In Stock Analysis





Inventory Value by Product

The inventory value analysis shows that Word Challenge holds the highest inventory value at approximately **₹6.3M**, followed by **Doctor Doll (₹4.8M)** and **Memory Learning Cards (₹4.3M)**. These products represent a significant portion of capital tied up in unsold inventory. High inventory value indicates that these products are occupying warehouse space and increasing carrying costs without generating equivalent sales. This suggests that some products are being stocked far above actual demand. Identifying these products is critical for reducing excess inventory and improving cash flow.

Units Sold vs Units In Stock Analysis

The comparison between units sold and units in stock clearly highlights several products with excess stock relative to their sales performance. Products such as **Word Challenge, Memory Learning Cards, Doctor Doll, Animal Puzzle, and Mystery Puzzle** have very high stock levels while selling comparatively fewer units. In contrast, some products such as **Coding Toy, STEM Kit, and Math Learning Kit** show lower stock levels with healthy sales, indicating stronger inventory efficiency. This imbalance suggests that certain slow-moving products are tying up capital, while fast-moving products may need tighter replenishment planning. The chart clearly separates underperforming inventory from better-managed products.

Overstock Products by Category

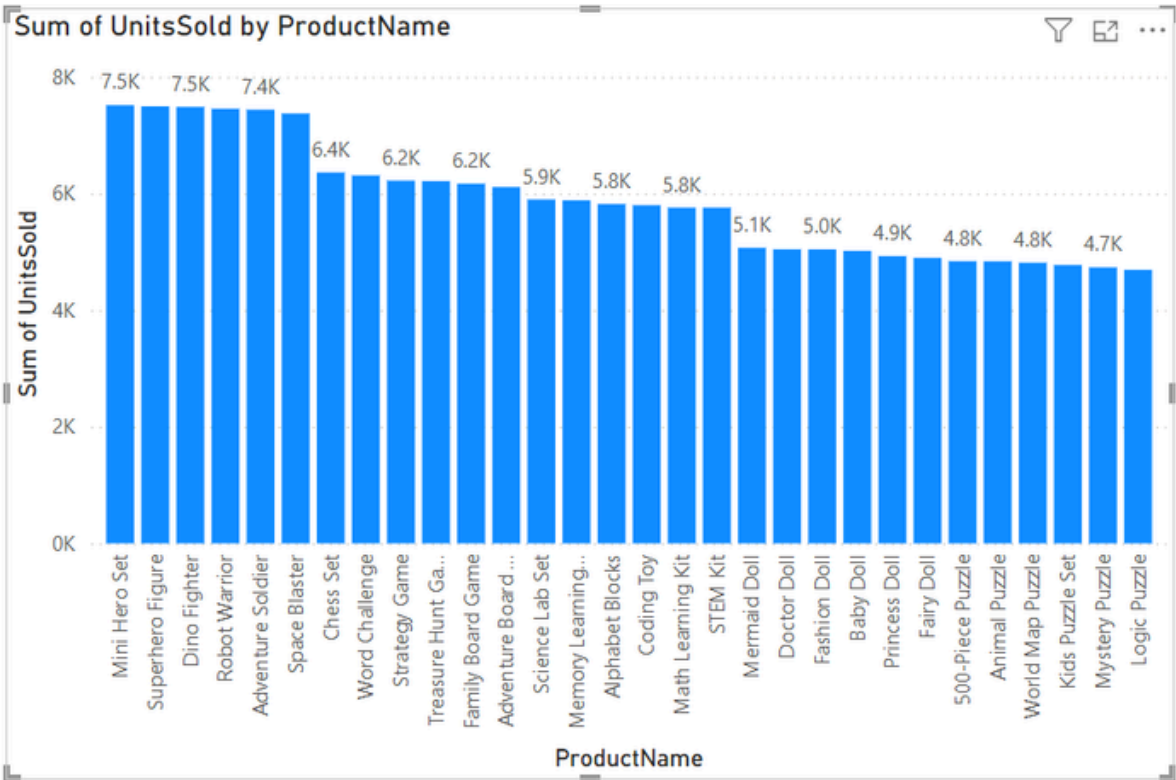
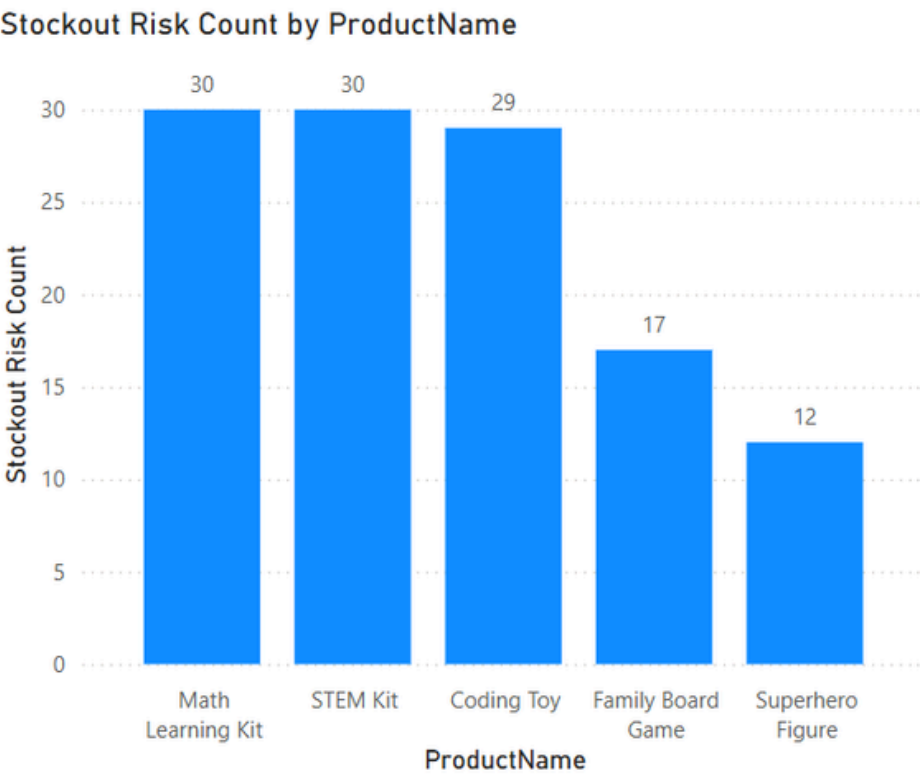
The overstock category analysis shows that Puzzles account for the highest share of overstock records (**869 records / 28.65%**), followed closely by Dolls (**807 / 26.61%**) and **Board Games (578 / 19.06%)**. This indicates that excess inventory is concentrated primarily in categories with slower or less predictable demand. Action Figures and Educational Toys contribute a smaller share of overstock, suggesting relatively better alignment between stock levels and sales demand. The results highlight which categories require immediate inventory corrections. Overstock in these categories is likely reducing warehouse efficiency and increasing holding costs.

Business Question Answer

Which products or categories have excess inventory that ties up capital?

The analysis shows that several products are carrying excess inventory, particularly Word Challenge, **Doctor Doll, Memory Learning Cards, Animal Puzzle, and Mystery Puzzle**, all of which hold high inventory value and stock levels compared to their sales volume. At the category level, **Puzzles and Dolls** contribute the largest share of overstock, making them the biggest drivers of excess inventory. These products and categories are tying up working capital, increasing storage costs, and reducing inventory efficiency. ToyJoy should prioritize inventory reduction strategies for these slow-moving items to free up warehouse capacity and improve cash flow.

07: Stockout Risk Analysis



Stockout Risk by Product

The stockout risk analysis highlights that **Math Learning Kit and STEM Kit have the highest stockout risk count at 30 instances each**, followed closely by **Coding Toy with 29 instances**. These products repeatedly reached critically low inventory levels, indicating that demand is stronger than current stocking levels. Family Board Game and Superhero Figure also show noticeable stockout occurrences, though at lower levels. The results suggest that several of ToyJoy's popular products are not being replenished quickly enough. These recurring stockout risks may be causing lost sales and missed revenue opportunities.

Units Sold by Product Analysis

The **units sold chart shows that Mini Hero Set, Superhero Figure, Dino Fighter, Robot Warrior, and Adventure Soldier are among the highest-selling products, each contributing around 7K-7.5K units sold**. This indicates strong customer demand for action figures and related products. **Educational products such as Math Learning Kit, STEM Kit, and Coding Toy also show healthy sales**, even though they are not at the very top of total units sold. The presence of high sales alongside repeated stockout risk suggests that some products are consistently understocked relative to demand. These products should be prioritized for tighter inventory planning and faster replenishment.

High Demand vs Low Stock Insight

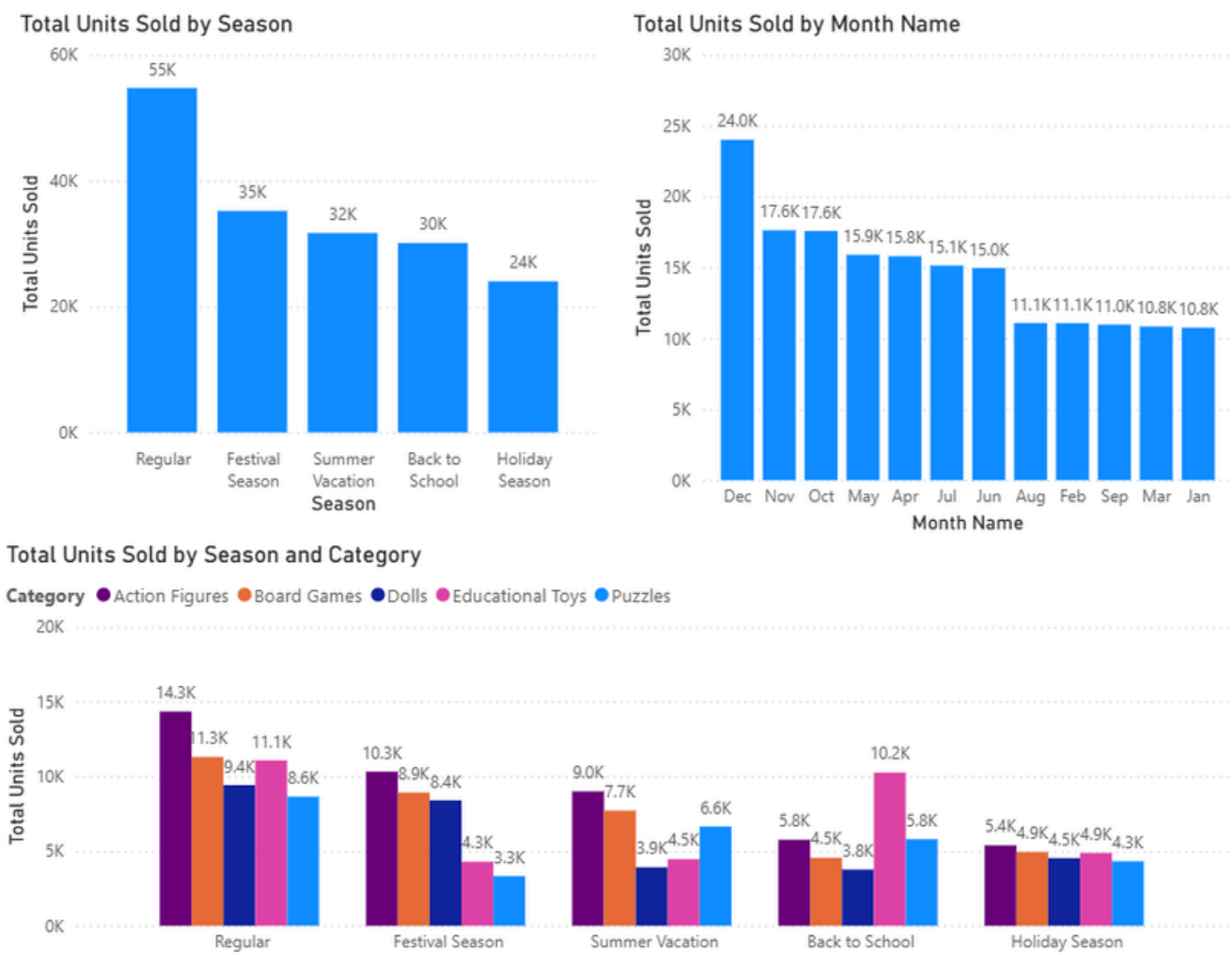
Combining stockout risk with units sold clearly shows that **Math Learning Kit, STEM Kit, Coding Toy, Family Board Game, and Superhero Figure are high-demand products facing repeated low-stock situations**. These products generate strong sales but frequently approach stockout levels, which means ToyJoy may be losing potential revenue whenever demand spikes. Educational toys and certain action figures appear especially vulnerable during seasonal demand peaks. This indicates that current reorder quantities or reorder timing are not sufficient for these fast-moving products. Improving replenishment planning for these items can directly reduce lost sales.

Business Question Answer

Which products are frequently out of stock, causing lost sales?

The analysis shows that **Math Learning Kit, STEM Kit, Coding Toy, Family Board Game, and Superhero Figure** are the products most frequently at risk of stockouts. These products combine repeated low-stock occurrences with strong sales demand, making them the most likely source of lost sales opportunities. Educational toys and selected action figures appear to be the most vulnerable to understocking, particularly during high-demand periods. ToyJoy should prioritize these products for faster replenishment, higher safety stock, and more accurate seasonal demand planning.

08: Seasonal Demand Analysis



Units Sold by Season Analysis

The seasonal sales analysis shows that the **Regular season contributed the highest sales at approximately 55K units, followed by Festival Season (35K) and Summer Vacation (32K). Back to School also generated strong demand with around 30K units sold, mainly driven by educational toys. Holiday Season recorded the lowest total units sold among the tagged seasons**, though it still represents an important demand period for specific toy categories. The results confirm that ToyJoy's demand is clearly influenced by seasonal shopping patterns and festival periods.

Units Sold by Month Analysis

The monthly sales chart shows that **December was the strongest month with approximately 24K units sold**, followed by **November and October with around 17.6K units each**. Months such as **May, April, July, and June also recorded strong sales**, reflecting the impact of summer vacation and school-related demand. In contrast, **January, March, August, and September show comparatively lower sales levels**. This pattern suggests that demand peaks are concentrated around year-end holidays, festival months, and school-related shopping periods. ToyJoy should align inventory build-up before these peak months to avoid stock shortages.

Category Sales by Season Analysis

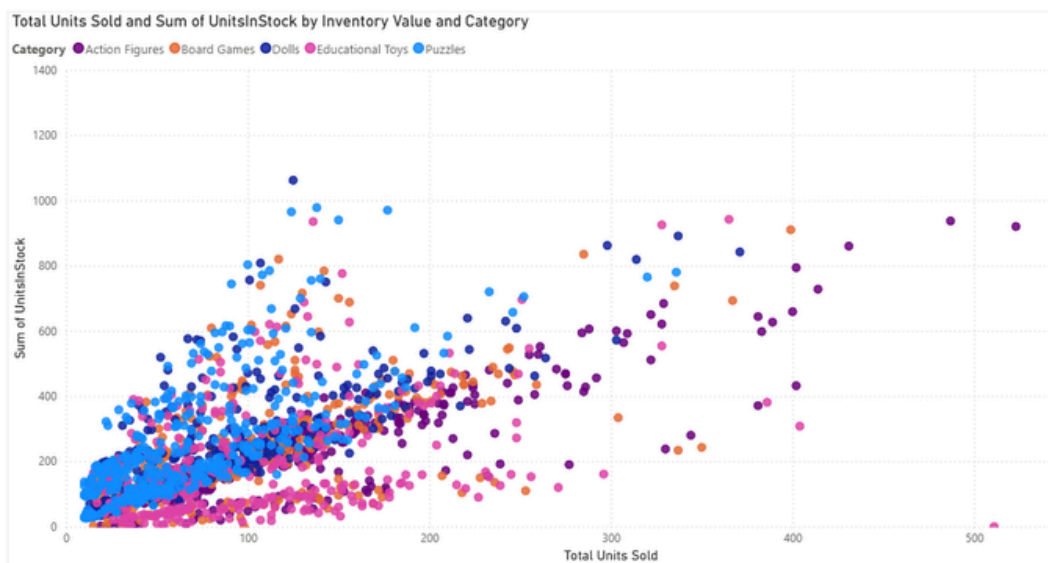
The **category-season chart highlights that Action Figures performed strongly across all seasons, especially during the Regular season and Festival Season. Educational Toys peaked significantly during Back to School, where they generated the highest category-level seasonal demand. Board Games and Dolls also showed stronger performance during the Regular and Festival seasons, while Puzzles recorded relatively stable but lower sales across most periods**. These differences show that each category follows a different seasonal demand pattern. Inventory planning should therefore be customized by category rather than using a single stocking approach for all toys.

Business Question Answer

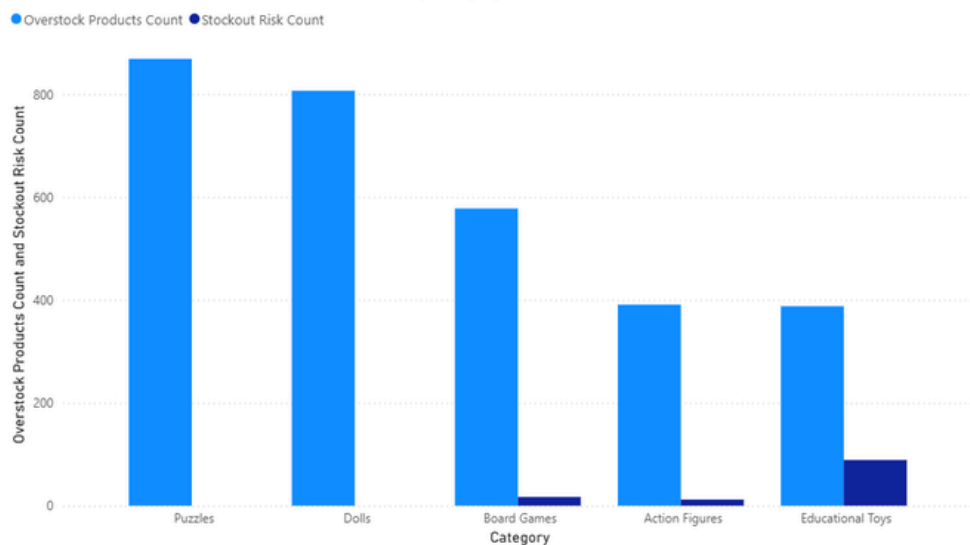
Can you identify seasonal trends in sales to guide stocking for the next period?

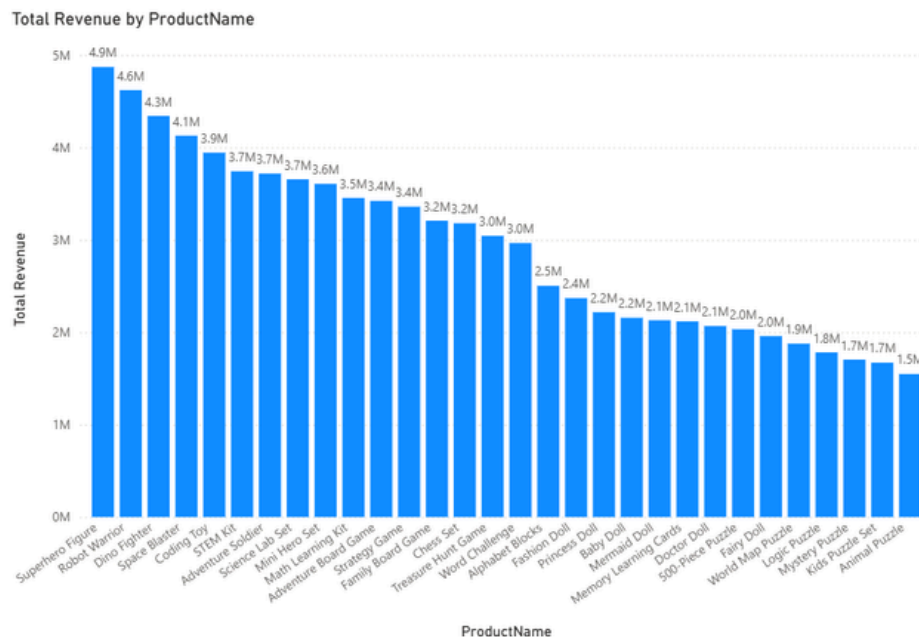
Yes, the analysis clearly shows that ToyJoy's sales are influenced by seasonal demand patterns. Festival Season, Summer Vacation, Back to School, and year-end holiday months all drive noticeable increases in sales. Educational Toys peak during the Back-to-School period, while Action Figures, Dolls, and Board Games perform strongly during festival and holiday periods. These patterns indicate that ToyJoy should increase inventory for specific categories ahead of peak demand months and reduce stock levels during slower periods. Seasonal planning will help the company avoid both stockouts and unnecessary overstock.

09: Inventory Adjustment & Reorder Strategy



Overstock Products Count and Stockout Risk Count by Category





Revenue by Product Analysis

The revenue chart shows that **Superhero Figure (₹4.9M)**, **Robot Warrior (₹4.6M)**, and **Dino Fighter (₹4.3M)** are the top revenue-generating products. **Products like Coding Toy and STEM Kit** also contribute strongly to total revenue. On the other hand, products such as **Animal Puzzle** and **Kids Puzzle Set** generate comparatively lower revenue. This helps identify which products should be prioritized and which ones need inventory reduction.

Inventory Value vs Units Sold Analysis

The scatter plot shows a clear difference between **slow-moving** and **fast-moving** products. **Puzzles and Dolls appear more frequently in the high-stock but lower-sales area**, indicating overstock. In contrast, Educational Toys and some Action Figures show stronger sales with tighter stock levels. This suggests that ToyJoy is overstocking slower products while some high-demand products need better replenishment.

Overstock vs Stockout by Category Analysis

The category comparison shows that **Puzzles and Dolls have the highest overstock counts**. **Educational Toys have the highest stockout risk**, while **Board Games show a mix of overstock and stockout pressure**. This means inventory issues differ by category and cannot be solved with one common stocking approach. ToyJoy needs category-wise inventory planning.

Business Question Answer

Recommend inventory adjustments for underperforming and high-demand products.

ToyJoy should reduce stock for slow-moving products in Puzzles and Dolls, especially products with **high stock and low revenue**. At the same time, high-demand products such as **Math Learning Kit, STEM Kit, Coding Toy, and Superhero** Figure should be stocked more aggressively. This will reduce blocked inventory while improving product availability for strong sellers.

Business Question Answer

Suggest strategies to optimize reorder points and quantities to reduce overstock and stockouts.

Reorder points should be increased for high-demand products, especially Educational Toys and top Action Figures. For slow-moving products, reorder quantities should be reduced and replenishment should happen less frequently. Seasonal demand should also be considered while planning stock for festivals, holidays, and Back-to-School periods. This will help ToyJoy maintain a better inventory balance.

10: Recommendations & Strategy

PAGE 9 – RECOMMENDATIONS & ACTION PLAN			
QUICK WINS		LONG-TERM STRATEGIES	
ACTION		STRATEGY	EXPECTED IMPACT
	Reduce stock of slow-moving products Focus on Word Challenge, Doctor Doll, Animal Puzzle, Mystery Puzzle and other low-selling items.		Build seasonal demand forecasting Use historical sales data to forecast demand for festival, summer and back-to-school periods.
	Increase stock for high-demand products Priority: Math Learning Kit, STEM Kit, Coding Toy, Superhero Figure and other fast movers.		Set category-wise reorder points Define reorder points based on sales velocity, lead time and stockout risk for each category.
	Create a priority restocking list Identify high-demand items and restock them on a priority basis.		Use ABC inventory classification Classify products by value and sales impact to focus on the most important items.
	Slow down reordering for low-selling Puzzles and Dolls Reduce order quantity and increase reorder interval for slow movers.		Track inventory turnover and stockout KPIs regularly Monitor KPIs such as inventory turnover, stockout count and days of inventory.
	Review stock weekly for top-selling seasonal products Monitor inventory for festival, holiday and back-to-school items.		Align purchasing with demand patterns Plan purchases based on sales trends, seasonality and product performance.
	Frees up warehouse space and reduces blocked capital.		Improves stock planning before demand peaks.
	Prevents stockouts and captures more sales.		Reduces both overstock and stockout risk.
	Improves product availability during peak demand.		Optimizes focus on high-value and high-demand items.
	Reduces excess inventory and carrying cost.		Supports data-driven decisions and continuous improvement.
	Helps react faster to demand spikes and trends.		Improves cash flow and overall inventory efficiency.

11: Final Conclusion & Business Impact

ToyJoy’s inventory challenges are mainly driven by overstock in slow-moving products and stockout risk in **high-demand products**. **Categories such as Puzzles and Dolls are carrying excess inventory, while Educational Toys and some Action Figures frequently face low-stock situations.** The analysis also shows clear seasonal demand patterns, with stronger **sales during festival periods, summer vacation, Back-to-School, and December.**

By reducing stock for underperforming products and increasing replenishment for fast-moving items, ToyJoy can improve inventory balance and reduce lost sales. Seasonal stocking, category-wise reorder points, and better monitoring of inventory KPIs will help the company lower carrying costs, improve product availability, and strengthen cash flow. Overall, a more demand-driven inventory strategy will improve efficiency, customer satisfaction, and business performance.